Hiw cos (-x) becomes cos c

You're asking how $\cos(-x)$ becomes $\cos x$.

## Simple explanation 👇

Cosine is an even function.

That means:

$$\cos(-x) = \cos(x)$$

## Why does this happen?

### 1️⃣ From the unit circle (easy to imagine)

- $\cos x$ represents the x-coordinate of a point on the unit circle.
- The angles $x$ and $-x$ are symmetric about the x-axis.
- Symmetric points have the same x-coordinate.

So:

$$\cos(-x) = \cos x$$

### 2️⃣ Using trigonometric identities

Cosine is defined using sine:

$$\cos(-x) = \sin\left(\frac{\pi}{2} - (-x)\right) = \sin\left(\frac{\pi}{2} + x\right) = \cos x$$

### 3️⃣ Even–odd rule (most important to remember)

- Cosine → Even
- Sine → Odd

So:

$$\cos(-x) = \cos x$$

$$\sin(-x) = -\sin x$$

## One-line memory tip 🧠

Cos ignores the minus sign.

If you want, I can also explain this using a graph or with a numerical example (like $\cos(-60^\circ)$).

Inverse of f(x) =log(x^2+1)

Let's find the inverse step by step in a clear and easy way.

Given function

$$f(x) = \log(x^2 + 1)$$

(Assume log means natural log or base 10 — the inverse process is the same.)

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## Step 1: Replace $f(x)$ with $y$

$$y = \log(x^2 + 1)$$

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## Step 2: Swap $x$ and $y$

$$x = \log(y^2 + 1)$$

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## Step 3: Remove the logarithm

Convert from log form to exponential form:

- If log is base 10:

$$10^x = y^2 + 1$$

- If log is natural log:

$$e^x = y^2 + 1$$

I'll continue with general form:

$$\text{antilog}(x) = y^2 + 1$$

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### Step 4: Solve for $y$

$$y^2 = \text{antilog}(x) - 1$$

$$y = \pm\sqrt{\text{antilog}(x) - 1}$$

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### ⚠️ Important: Inverse exists only if function is one-to-one

Since $x^2$ is not one-to-one, we must restrict the domain.

Choose domain:

- If $x \geq 0$ → take positive root
- If $x \leq 0$ → take negative root

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### ✅ Final inverse function

If domain is $x \geq 0$:

$$f^{-1}(x) = \sqrt{\text{antilog}(x) - 1}$$

If domain is $x \leq 0$:

$$f^{-1}(x) = -\sqrt{\text{antilog}(x) - 1}$$

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### One-line summary 🧠

> Inverse of $\log(x^2 + 1)$ involves square root, so domain restriction is necessary.

If you want, I can also:

- Verify by composition $f(f^{-1}(x))$
- Explain why domain restriction is required

- Solve using graph idea

Just tell me 👍

Why negative sign removes ftom cos x

Good question — this is a very common confusion 👍

Let's explain it clearly and simply.

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## Why does the negative sign disappear in $\cos(-x)$?

### Short answer 🧠

Because cosine is an even function.

$$\boxed{\cos(-x) = \cos x}$$

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## What does "even function" mean?

A function is even if:

$$f(-x) = f(x)$$

Cosine satisfies this rule, so the minus sign has no effect.

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## Intuitive explanation (unit circle) 🔵

- $\cos x$ = x-coordinate on the unit circle
- Angles $x$ and $-x$ are mirror images across the x-axis
- Mirror points have the same x-coordinate

So:

$$\cos(-x) = \cos x$$

## Graph explanation 📈

- The graph of $\cos x$ is symmetric about the y-axis
- This symmetry means:

$$\cos(-x) = \cos x$$

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## Compare with sine (to avoid confusion)

- Cosine → even → minus disappears
- Sine → odd → minus stays

$$\sin(-x) = -\sin x$$

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## One-line memory rule ✍️

> Cos keeps the value, sine changes the sign.

If you want, I can also show this using a numerical example like
$\cos(-60^\circ) = \cos(60^\circ)$ or explain it using series expansion.

Like x^2 we know for negative input it outputs the  positive response so how we relate it with cos x

Excellent connection 👍

You're thinking in exactly the right way.

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## How is $\cos x$ like $x^2$?

### 1️⃣ Property of $x^2$

For any number $x$:

$$(-x)^2 = x^2$$

So:

$$f(-x) = f(x)$$

That's why $x^2$ is an even function.

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### 2️⃣ Same idea applies to cosine

For cosine:

$$\cos(-x) = \cos x$$

So cosine is also an even function, just like $x^2$.

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## Why does this happen for cosine?

### 🔵 Unit circle explanation (key idea)

- $\cos x$ is the horizontal (x) coordinate of a point on the unit circle.
- When you go to angle $-x$, the point reflects across the x-axis.
- Reflection across the x-axis does not change the x-coordinate.

So:

$$\cos(-x) = \cos x$$

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## Algebraic comparison (very important)

| Function | Result for $-x$ | Reason |
|---|---|---|
| $x^2$ | $(-x)^2 = x^2$ | Squaring removes sign |
| $\cos x$ | $\cos(-x) = \cos x$ | Symmetry of unit circle |
| $\sin x$ | $\sin(-x) = -\sin x$ | y-coordinate changes sign |

## One-line intuition 🧠

> $x^2$ ignores the sign because of squaring;
> cosine ignores the sign because of symmetry.

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## Final takeaway

- $x^2$ → algebraic evenness
- $\cos x$ → geometric evenness
- Same rule:

$$f(-x) = f(x)$$

If you want, I can also explain this using the graph of cos x or a Taylor series connection with $x^2$.

What means symmetry

## What does symmetry mean? (very simple)

Symmetry means:

> Both sides look the same when you reflect or fold something.

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## Everyday examples 🌸

- 🦋 A butterfly: left side = right side
- 🪞 Mirror image: your face looks the same on both sides
- 📄 Fold a paper in half → both parts match

That matching is called symmetry.

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## Symmetry in mathematics

## 1️⃣ Symmetry about the y-axis (important here)

A graph is symmetric about the y-axis if:

- Left side of the graph = right side

Mathematically:

$$f(-x) = f(x)$$

Such functions are called even functions.

Examples:

- $f(x) = x^2$
- $f(x) = \cos x$

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## 2️⃣ Symmetry about the origin

A graph is symmetric about the origin if:

- Rotating it 180° gives the same graph

Mathematically:

$$f(-x) = -f(x)$$

Such functions are called odd functions.

Examples:

- $f(x) = x$
- $f(x) = \sin x$

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## Visual idea (imagine this)

- For $x^2$: point at (2,4) ↔ point at (−2,4)
- For $\cos x$: value at $x$ = value at $-x$

That matching is symmetry.

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## One-line definition 🧠

> Symmetry means the shape or graph remains unchanged when reflected.

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### Connect to your earlier question

- $x^2$ → symmetric → output same for ±x
- $\cos x$ → symmetric → output same for ±x

So:

$$\cos(-x) = \cos x$$

If you want, I can also explain symmetry using graphs, tables, or real-life analogies 😊



What means symmety avout x axis

### What does symmetry about the x-axis mean? (very easy)

Symmetry about the x-axis means:

> If a point is above the x-axis, an identical point exists the same distance below the x-axis.

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### Simple idea (mirror concept) 🪞

Imagine the x-axis is a mirror:

- Point $(x, y)$
- Mirror point → $(x, -y)$

If both points are on the graph, the graph is symmetric about the x-axis.

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### Mathematical condition

A graph is symmetric about the x-axis if:

$$(x, y) \text{ is on the graph } \Rightarrow (x, -y) \text{ is also on the graph}$$

## Important note ⚠️

👉 Most functions cannot be symmetric about the x-axis

(because one $x$ would give two $y$-values).

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## Examples

### ✔ Symmetric about x-axis

- $x^2 + y^2 = 4$ (circle)
- $y^2 = x$ (sideways parabola)

### ❌ NOT symmetric about x-axis

- $y = x^2$
- $y = \cos x$

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## Compare with other symmetries

| Type | Condition |
|---|---|
| x-axis symmetry | $(x, y) \to (x, -y)$ |
| y-axis symmetry | $f(-x) = f(x)$ |
| origin symmetry | $f(-x) = -f(x)$ |

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## One-line memory rule 🧠

> x-axis symmetry flips y, not x.

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## Link to cosine (your earlier topic)

- $\cos x$ is symmetric about the y-axis, not the x-axis
- That's why:

$$\cos(-x) = \cos x$$

If you want, I can draw a simple table or graph explanation to make this even clearer 😊

Thanks

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You're welcome! 😄

If you want, I can also make a small visual diagram showing cosine symmetry, $x^2$ symmetry, and axis types — it makes it super easy to remember. Do you want me to do that?